

Bioinformatics Analysis Workflow



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Third-generation data:

The third-generation sequencing instrument Nanopore, which uses single-molecule sequencing technology, has the advantage of ultra-long read lengths and plays an important role in the assembly of complete genomes. The raw sequencing data is saved in fast5 format files, which contain sequencing sequences, base quality values, and other information. By performing quality control on the raw sequencing data and removing low-quality ($Q < 7$) sequences, clean data that can be used for analysis is obtained. Statistical analysis is performed on the clean data to obtain information such as the total amount of data, read lengths, and quality value distributions.

Second-generation data:

The raw data obtained from sequencing (Raw Data) may contain a certain proportion of low-quality data. In order to ensure the accuracy and reliability of subsequent information analysis results, the raw data needs to be filtered to obtain clean data (Clean Data). The specific processing steps are as follows:

Remove reads that have a proportion of low-quality bases (quality value ≤ 38) exceeding a certain threshold (default set to 40bp).

Remove reads that have a proportion of N bases exceeding a certain threshold (default set to 10bp).

Remove reads that have an overlap with the adapter exceeding a certain threshold (default set to 15bp).

For projects with small genomes or potential host contamination, the reads that may originate from the host should be filtered out by aligning them with the host database.

To assemble the genome, the Flye software (<https://github.com/fenderglass/Flye>, version 2.9-b1768) is used to obtain preliminary assembly results. Then, the Racon software (version 1.4.13) is used to perform three rounds of error correction based on third-generation sequencing data on the assembly results. After that, three rounds of error correction are done using the Pilon software (version 1.22) based on second-generation reads, resulting in the final assembly result.

Genome Component Analysis

Coding Genes

Starting from the final assembly results (≥ 500 bp) of each sample, ORF (Open Reading Frame) prediction and filtering are performed using the corresponding software. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component/Gene".

For gene prediction in fungal samples, the following methods are used:

- (1) If transcriptome data is provided, de novo prediction is performed using Transdecoder/Glimmer/Snap based on the transcriptome data, as well as from transcriptome data using Cufflinks, de novo prediction using Augustus (version 2.7), and homology-based prediction using Genewise (version 2.4.1) with reference sequences (if provided). The results from multiple methods are then integrated using EVM and validated with a second round of PASA.
- (2) If transcriptome data is not provided but reference sequences are available, homology-based prediction is performed using Genewise.
- (3) If neither transcriptome data nor reference sequences are provided, de novo prediction is performed using Augustus.

Based on the distribution of repetitive sequences in the genome, they can be divided into two major categories: dispersed repeat sequences and tandem repeat sequences. Dispersed repeat sequences are a type of repeat sequence that differs in organization from tandem repeat sequences, and they are scattered throughout the genome. Tandem repeat sequences (TR) are repetitive sequences that consist of adjacent, repeated patterns of a specific nucleotide sequence, repeated two or more times. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component/Repeat".

The prediction of dispersed repeat sequences is performed using the RepeatMasker software (version 4.0.5), while the TRF (Tandem Repeats Finder, version 4.07b) is used to search for tandem repeat sequences in the DNA sequences.

Non-coding RNA (ncRNA)

Non-coding RNA (ncRNA) is a class of RNA molecules that are widely present in small genomes, bacteria, archaea, and eukaryotes, and they perform various biological functions. Unlike coding RNA, ncRNA does not carry information for protein translation and directly acts on biological processes at the RNA level. For microorganisms, common types of ncRNA include sRNA, rRNA, tRNA, snRNA, snoRNA, and miRNA, among others. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component/ncRNA".

Common methods for predicting different types of ncRNA are as follows:

tRNA: The tRNAscan-SE software can be used to predict tRNA regions and their secondary structures.

rRNA: By aligning against an rRNA database, rRNAs that are closely related to the database (with a default identity of $\geq 50\%$) can be identified. Additionally, the rRNAmmer software can be used to predict newly identified and

sRNA: First, the Rfam software is used for Rfam database annotation. Then, the cmsearch program (version 1.1rc4) is used to determine the final sRNA predictions.

Gene function analysis

Gene function annotation

There are several commonly used functional databases for annotation, including GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), COG (Cluster of Orthologous Groups of proteins), KOG, eggNOG (evolutionary genealogy of genes: Non-supervised Orthologous Groups), NR (Non-Redundant Protein Database), TCDB (Transporter Classification Database), Swiss-Prot, and CAZy (Carbohydrate-Active enZymes Database). The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/General_Gene_Annotation" for each database.

The basic steps for functional annotation are as follows:

Perform a BLAST comparison (blastp) between the predicted genes and each functional database (with an e-value threshold of $\leq 1e-5$).

Filter the BLAST results: For each sequence, select the highest-scoring alignment result (default criteria: identity $\geq 40\%$, coverage $\geq 40\%$) for annotation.

Effector

Secretory proteins

Secretory proteins are proteins that are synthesized inside cells and then secreted outside the cells to exert their functions, guided by signal peptides. Many important enzymes necessary for life activities are found in secretory

proteins. The N-terminal of secretory proteins is a signal peptide composed of 15-30 amino acids, which plays a key role in protein secretion. The SignalP tool can be used for signal peptide prediction to detect the presence of a signal peptide and transmembrane regions, and to predict whether a protein sequence is a secretory protein. The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/Secretory_Protein".

Cellular cytochrome P450 database annotation

Cytochrome P450 (cytochrome P450 or CYP450, abbreviated as CYP450) is a superfamily of heme-thiolate proteins. It is involved in the metabolism of endogenous substances and exogenous substances, including drugs and environmental compounds.

Results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/P450".

Analysis of secondary metabolism gene clusters

Secondary metabolites are synthesized by microorganisms from primary metabolites during a specific growth phase. They do not have a clear function in the microbial life activities and are not essential for growth and reproduction. The antiSMASH program (version 2.0.2) was used to predict the genome for secondary metabolite gene clusters.

Results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/Secondary_Metabolism".

Pathogenicity analysis

For fungi, there are currently annotated databases for pathogenicity and drug resistance, including PHI (Pathogen Host Interactions Database) and DFVF (Database of Fungal Virulence). Results can be found in

"04.Genome_Function.tar.gz" or in the "Pathogenicity" folder within "04.Genome_Function" which contains various databases.

Comparative Genomics Analysis

Whole-genome collinearity analysis

Collinearity refers to the gene linkage relationship in genetics, which is the phenomenon where homologous genes on different species' genomes are arranged in the same order. The degree of collinearity between two species can serve as a scale to measure their evolutionary distance and determine their phylogenetic relationship. Results can be found in "05.Comparative_Genomics.tar.gz" or "Syteny" in "05.Comparative_Genomics".

SNP Analysis

SNP (Single Nucleotide Polymorphism) refers to DNA sequence polymorphism caused by single nucleotide variations at the genomic level, including single base substitutions, insertions, and deletions.

MUMmer software (version 3.22) was used for individual SNP detection, and SNP functionality was annotated based on the positional relationship and interaction between SNPs and genes. Results can be found in "05.Comparative_Genomics.tar.gz" or "SNP" in "05.Comparative_Genomics".

InDel Analysis

InDel refers to the insertion and deletion of small fragments in the genome. The LASTZ software was used to detect small fragment insertions and deletions (InDels) with a length of less than 50bp. The results can be found in "05.Comparative_Genomics.tar.gz" or "InDel" in "05.Comparative_Genomics".

Structural Variation (SV) refers to large-scale insertions, deletions, inversions, translocations, and other sequence changes at the genomic level.

To analyze SV, the MUMmer software (version 3.22) is used for fast alignment of two genomes to determine the relative orientation of sequences and adjust the genome by aligning the starting points. The Lastz software (version 1.02.00) is used for whole-genome alignment. Software packages like chainNet are used to connect scattered alignment results into longer alignment results in formats such as chain and net. Based on the new arrangement of alignment blocks and their relative orientations, alignment blocks representing translocations, inversions, and translocations with inversions are identified. The SV regions between adjacent alignment blocks on the two genomes are determined based on their distance relationships.

The results can be found in "05.Comparative_Genomics.tar.gz" or "SV" in "05.Comparative_Genomics".

Population Evolution Analysis

Analysis of Shared and Specific Genes

Genes that are present in all samples are referred to as core genes, while genes that are not shared among all samples are called dispensable genes. Specific genes are genes that are unique to a particular sample. The combination of all dispensable genes and core genes is referred to as the pan-gene. The core genes and specific genes are likely to correspond to common characteristics and specific traits of the samples, and can be used as a basis for studying functional differences between samples.

The results can be found in "05.Comparative_Genomics.tar.gz" or "Core-Pan" in "05.Comparative_Genomics".

In genomic evolution, a gene can produce two or more copies through gene duplication, which then form a gene family. A gene family is a group of genes that share a common ancestor, and the different genes within the family often have similar structures and functions. Since gene families are formed through species divergence, studying gene family evolution can reveal the evolutionary history of genes, gene functions, and species divergence, providing evidence and clues for predicting unknown gene functions and studying gene evolutionary history.

To analyze gene families, the BLAST software is used to pairwise compare the proteins of multiple target genomes. The unreliable alignment results are filtered, and then redundancy is removed using Solar. Hcluster-sg is used to cluster proteins based on their alignment similarities, resulting in gene family clustering. The results can be found in "05.Comparative_Genomics.tar.gz" or "Gene_Family" in "05.Comparative_Genomics".

Species Evolution Analysis

A phylogenetic tree, also known as an evolutionary tree, is a tree that depicts the evolutionary relationships between species, indicating their common ancestors. It is used to represent the results of phylogenetic analysis and describe the evolutionary relationships between species.

To analyze species evolution, the Neighbor-Joining (NJ) method is used to construct a phylogenetic tree. The results can be found in "05.Comparative_Genomics.tar.gz" or "Phylogenetic_Tree" in "05.Comparative_Genomics".